

PlantCare (EcoVision): Intelligent Plant Disease Classification Platform

Final Demonstration & Project Submission Presentation

C0543/C05430

Computer Vision Project Assignment (Application Track)

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THE PROBLEM: DISEASE COSTS FARMERS MORE THAN YIELDS

Phytopathological infections destroy **20% to 40%** of global crop yields annually.

Smallholder farmers lack immediate access to trained agronomists and plant extension officers.

Delayed or inaccurate diagnosis drives improper pesticide use, rising costs and environmental harm.

Automated folia diagnostics on commodity smartphones deliver **sub-second** neural inference for early intervention.



PROJECT OBJECTIVES & SYSTEM SCOPE

Stronger AI Benchmarking

Compare ResNet18 (baseline) vs. MobileNetV3-Large across 38 crop disease and healthy leaf classes.

Modern, Decoupled Architecture

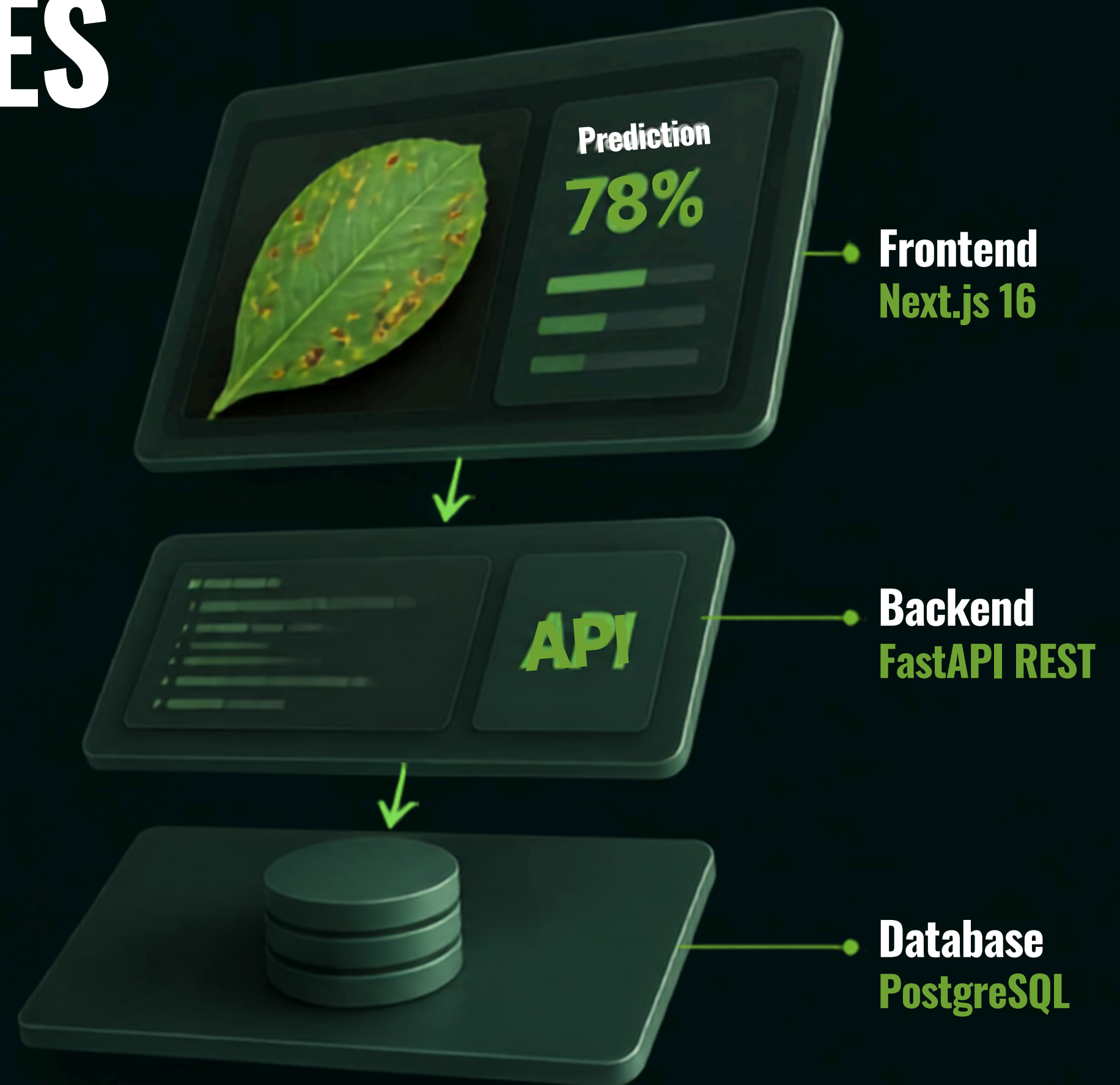
An asynchronous web system connecting Next.js 16 frontend, FastAPI REST backend, and PostgreSQL.

Actionable Guidance at the Point of Need

Pair prediction probabilities with clear treatment protocols and prevention rules.

Robust in Real Conditions

Evaluate performance under lesion ambiguity, background noise, and challenging test splits.



DATASET & PREPROCESSING PIPELINE

~87,000 RGB images

38 classes across 14 crop species

Data splits

Training ~70% (61,000) | Validation ~20% (15,000)

Test evaluation 10,703 images

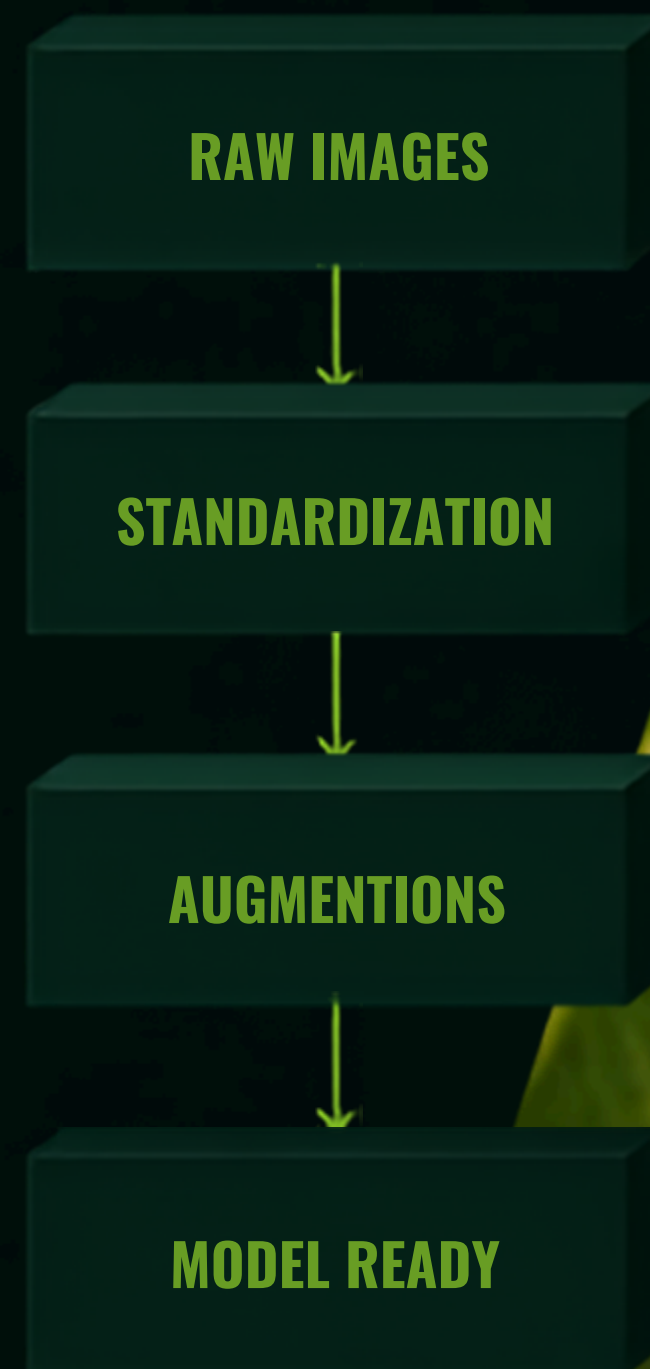
Preprocessing pipeline

Resized to 224×224 , PyTorch FloatTensors,
normalized with ImageNet stats

Data augmentations

Random Horizontal Flip ($p=0.5$)

Random Rotation ($\pm 30^\circ$), Random Affine zoom
& Brightness Jitter (± 0.3)



COMPUTER VISION ARCHITECTURE COMPARISON

BASELINE MODEL (RESNET18)



18-layer residual network with shortcut connections

Linear classifier (512 → 38 classes)

~44 MB model size

~45 ms CPU inference latency per image

PROPOSED MODEL (MOBILENETV3-LARGE)



Depthwise separable convolutions with inverted residual blocks and SE attention

~16 MB model size (63.6% footprint reduction)

~14 ms CPU latency (3.5x speedup)

99.57% | identical test evaluation accuracy

FULL-STACK MICROSERVICES ARCHITECTURE



NEXT.JS 16 WEB CLIENT

Built with React 19, TypeScript, and Tailwind CSS v4. Client-side image compression (browser-image-compression) shrinks uploads to $\pm 1\text{MB}$ in $<150\text{ms}$.



FASTAPI REST BACKEND

Asynchronous Python 3.12 server with Unicorn. Validates upload MIME types (JPEG, PNG, WEBP), serves UUID static images, and runs PyTorch inference.



POSTGRESQL DATABASE STORE

Sync SQLAlchemy 2.0 ORM manages 26 seeded disease records (description, causes, treatments, prevention) and user scan history logs.



DOCKER COMPOSITION

Multi-contained orchestration (db, backend, frontend) ensures environment reproducibility.

Quantitative Experimental Results

99.57% **Overall Accuracy**
10,657 / 10,703 images correct

1.00 **Weighted F1-Score**
Precision: 1.00 | Recall: 1.00

16 MB **Model File Size**
63.6% reduction vs. ResNet18

14 ms **CPU Inference Latency**
3.5x speedup

CROP-WISE ACCURACY BREAKDOWN

Apple Scab, Rot, Rust, Healthy	1,943 images F1: 1.00	99.8%
Grape Rot, Esca, Blight, Healthy	1,805 images F1: 1.00	100.0%
Potato Early/Late Blight, Healthy	1,426 images F1: 1.00	99.7%
Pepper Bell Bacterial Spot, Healthy	975 images F1: 1.00	99.8%
Peach & Orange Spot, Greening	1,394 images F1: 1.00	100.0%
Corn Gray Spot, Common Rust, Blight	1,829 images F1: 0.98	98.5%

Qualitative Results & Live Software Walkthrough

01

Scan & Client Com compression

Drag-and-drop / mobile camera upload. Com compresses 6MB high-res photo to <400KB in <150ms.

02

Diagnostic Results View

Animated confidence bar (e.g. 94.0%) and severity badges (Low, Moderate, Severe) returned in 220ms API response time.

03

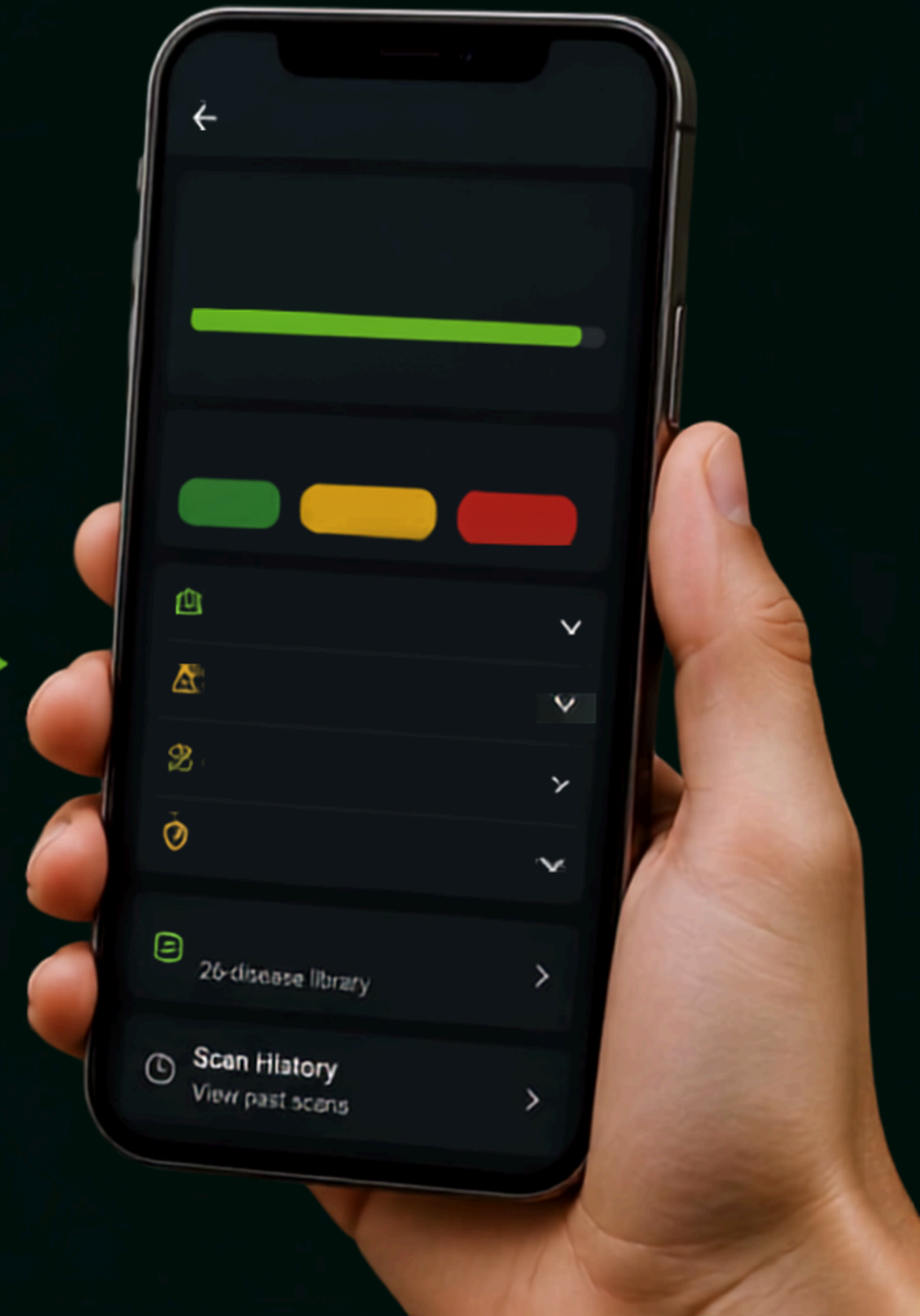
Actionable Recommendation Accordion

RecommendationPanel displaying Description, Root Causes, Step-by-Step Treatment, and Prevention rules.

04

Knowledge Base & History

Searchable 26-disease library grid and local scan history log for crop tracking.



FAILURE CASE & LIMITATION ANALYSIS

Corn Gray Leaf Spot recorded a lower **recall of 0.95** due to rectangular tan lesion overlap with Northern Leaf Blight under shadowed lighting.

Unweighted Macro F1 was **0.88** because 15 out of 38 trained classes were unrepresented in the test folder split.

Clean lab photos achieve **>99%** confidence, but field images with soil, weeds, or hands drop confidence **below 60%**.



INDIVIDUAL CONTRIBUTIONS & CONCLUSION

25%	Dulaj Ashen (E/22/032) PyTorch Modeling Pipeline, MobileNetV3 Optimization, evaluate.py Scripting, GitHub Management.
25%	Minhaj Ali (E/22/237) Next.js 16 Frontend Architecture, RecommendationPanel UI, Upload Flow, Client Image ComPRESSION, Documentation.
25%	Tharaka Denuwan (E/22/232) FastAPI Backend Microservice, REST Endpoint Routing (/predict, /diseases, /history), Static Media Handler.
25%	Rinushan Gunawardana (E/22/124) PostgreSQL Schema Design, Async SQLAlchemy ORM, seed_diseases.py Knowledge Base Scripting, Docker Composition.

IN SUMMARY

Successful deployment of a high-accuracy, lightweight plant disease classification platform supporting actionable guidance for farmers.

GITHUB REPOSITORY

<https://github.com/DulajE032/plantCare.git>

**THANK YOU
FOR YOUR ATTENTION.**

